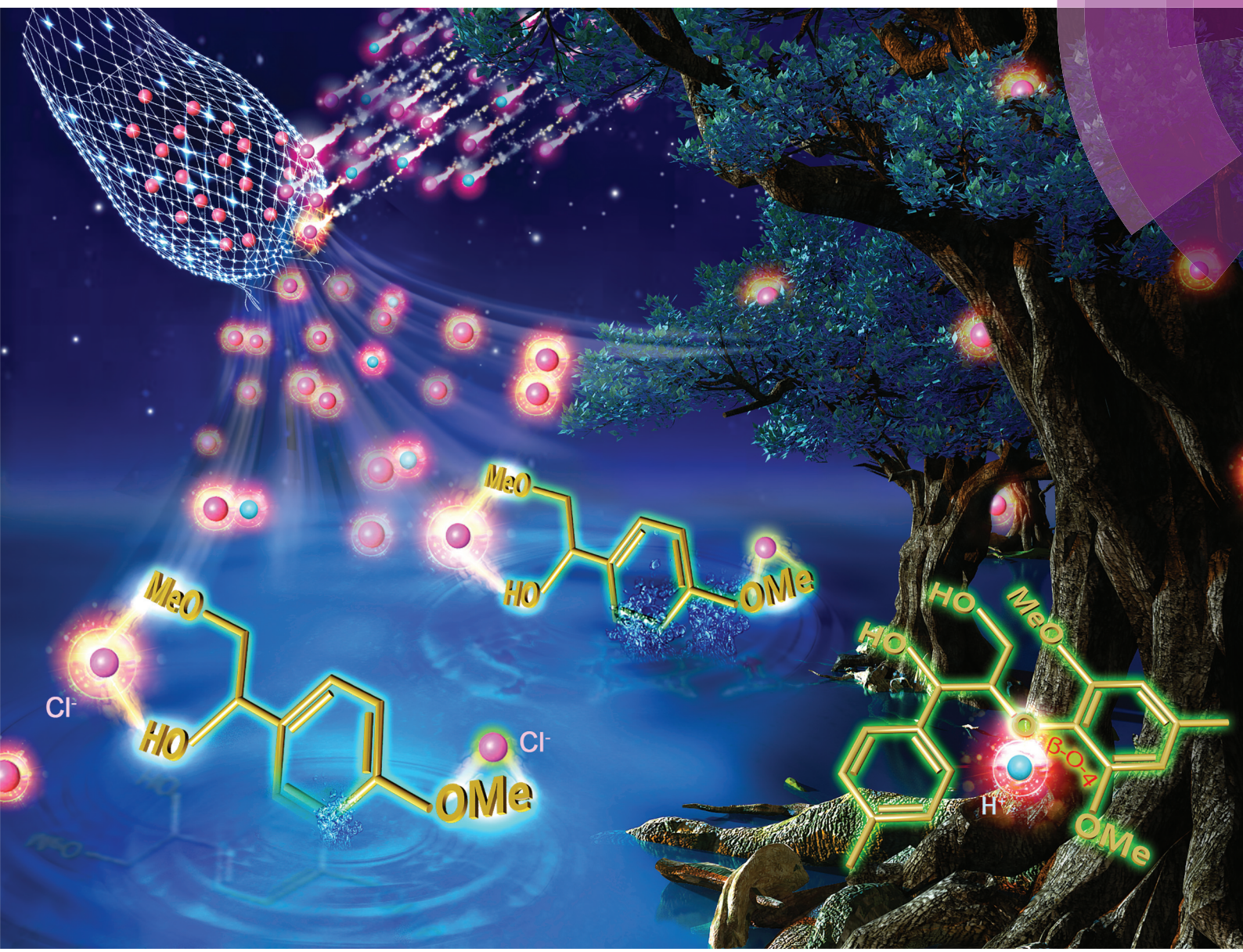
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Multiple hydrogen bond coordination in three-constituent deep eutectic solvents enhances lignin fractionation from biomass†

Qinqin Xia, Yongzhuang Liu, Juan Meng, Wanke Cheng, Wenshuai Chen, Shouxin Liu, Yixing Liu, Jian Li and Haipeng Yu *

As an emerging generation of green solvents, deep eutectic solvents (DESs) are promising for the pretreatment of lignocellulose and the production of biochemicals. However, not all DESs are effective for the cleavage of lignin–carbohydrate complexes (LCCs) in lignocellulose and the fractionation of lignin. In this study, we analyzed the nature of complex molecular interactions between choline chloride (ChCl) and glycerol in ChCl/glycerol (1 : 2) DES using density functional theory and the Kamlet–Taft solvatochromic method. The ChCl–glycerol DES exhibited weak competing interactions towards the linkages in the LCC network because the intramolecular hydrogen bonds (H-bonds) in ChCl–glycerol were constrained by mutually anionic H-bonds ($[Cl(glycerol)]^-$) and cationic H-bonds ($[Ch(glycerol)]^+$). Furthermore, because of the absence of active protons and acidic sites, the DES was unable to cleave ether bond linkages in the LCCs. Accordingly, we designed a three-constituent DES (3c-DES) by coordinating $AlCl_3{\cdot}6H_2O$ in ChCl/glycerol DES based on an acidic multisite coordination theory. The competition of anionic H-bonds and unidentate aluminum ligands was synchronized to form supramolecular complexes, allowing the multisite bridging ligands to cleave both the H-bonds and ether bonds in LCCs. Consequently, the lignin fractionation efficiency was significantly improved from 3.61% to 95.46%, and the lignin purity reached 94 ± 0.45%.

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With the depletion of fossil energy resources, biomass utilization is of significant importance for the production of alternative renewables.[1] Recently, the biomass pretreatment efficiency has been improved with the development and better understanding of novel solvents that do not harm the environment and are compatible with nature.[2] Deep eutectic solvents (DESs) have emerged as a promising new generation of green solvents which fully accord with the 12 principles of green chemistry. They are inexpensive, easy to prepare, nontoxic, biodegradable, and environmentally friendly.[3] Compared with toxic solvents and expensive solvents, DESs can be used in lieu of conventional chemical solvents and have a low ecological footprint.

Some acidic DESs that show an efficient lignin extraction ability have been reported.[4] However, not all types of DESs can be applied to biomass fractionation. Glycerol, as a natural compound and the main by-product generated by the biodiesel industry, has been assessed for use as a sustainable solvent for green chemistry by Gu and Jérôme.[5] Glycerol is also an inexpensive starting material and a kind of hydrogen bond donor (HBD) constituent for synthesizing glycerol-based DESs. Glycerol-based DESs have been prepared as easily accessible, biodegradable, and inexpensive alternatives to conventional aprotic cation–anion paired ionic liquids.[6] Glycerol-based DESs displayed excellent fluidity coupled with high thermal stability, and are effective in the electrochemical deposition of metal films or the extraction of active ingredients.[7] However, glycerol-based DESs have been frequently reported to be less effective for biomass fractionation. For example, a choline chloride (ChCl) and glycerol (ChCl/glycerol) DES was used to dissolve corncobs at 115 °C for 15 h, but it only showed a low lignin solubility of 4.4%.[8] Alvarez-Vasco *et al.* reported that the removal of lignin in Douglas fir and poplar was difficult after pretreatment with ChCl/glycerol DES.[9] Fang *et al.* also reported that ChCl/glycerol DES had little effect on removing the lignin from date palm residues.[10] It is still not well understood how the nature of the interactions between the organic matter and the ChCl/glycerol DES affect the biomass fractionation.[11] Therefore, an in-depth study is needed to explain the mechanism of poor biomass fractionation efficiency.

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Computational studies and theoretical parameters are important to predict the chemical properties of molecular solvents. For example, several empirical and semi-empirical polarity scales have been used to predict the solubility of lignocellulosic biopolymers or the pretreatment efficiency, including the Kamlet–Taft (K–T) solvatochromic method,[12] conductor-like screening models for real solvents,[13] and Hansen solubility parameters.[14] However, empirical prediction models have some limitations, as they can only predict the results according to the measured data. With the development of computational molecular science, molecular dynamics simulations and quantum calculations have been used to gain a better understanding of the exact mechanism of lignocellulose fractionation in solvents. For example, molecular dynamic simulations have been used to reveal how tetrahydrofuran/water co-solvents interact with the lignocellulose components and contribute to the pretreatment process.[15] The mechanism of the excellent solubility of lignin in ethylene glycol was explained through the quantum calculation of lignin model compounds.[16] Recently, quantum calculation analyses have also been conducted on ChCl-based DESs. Wagle *et al.* reported different interactions and thermodynamics in three types of ChCl-based DESs. The physical properties could be changed by tuning the choline–Cl^- interaction with HBDs.[17] García *et al.* studied the relationship between the melting point and hydrogen bond (H-bond) networks in 45 types of DESs by a density functional theory (DFT) study and Bader's theory.[18] Li *et al.* systematically studied the nature of the interactions between ChCl/glycerol DES and SO_2 using DFT calculations. The results showed that SO_2 interacted with both the hydrogen bond acceptor (HBA) and HBD through charge-transfer interactions.[19] However, to the best of our knowledge, no theoretical study has been performed on the biomass fractionation properties of ChCl/glycerol DES.

Herein, we try to obtain an in-depth understanding of the intermolecular interactions in ChCl–glycerol DES by studying them through a combination of DFT electronic structure calculations and the K–T solvatochromic parameters. Specifically, we focus on the molecular geometries, charge delocalization effects, and thermochemistry of these systems, exploring a better understanding of "What is important for lignocellulose fractionation using glycerol-based DES?" On the basis of the results, we aim to make full use of the advantage of the designable diversity of DESs to develop more efficient three-constituent DESs (3c-DESs) for lignin fractionation. 3c-DESs are systematically characterized by several spectroscopic approaches and the K–T solvatochromic parameters, and their lignin extraction ability is tested by applying to poplar wood under different conditions (Scheme 1).

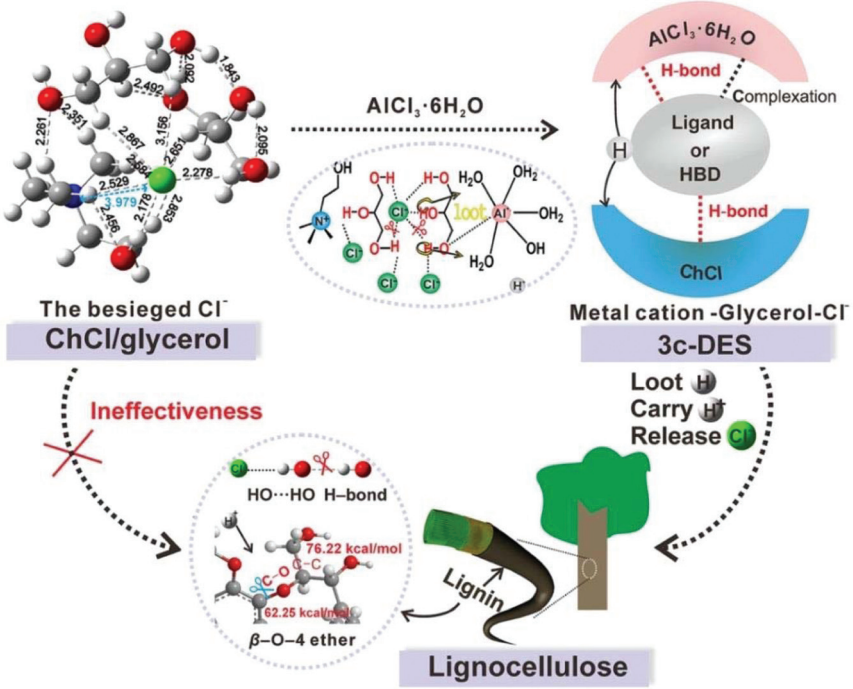
Experimental

Preparation of ChCl/glycerol DES

ChCl and glycerol at a molar ratio of 1 : 2 were mixed and stirred at 60 °C until a transparent liquid was obtained.

Preparation of ChCl/glycerol/$AlCl_3{\cdot}6H_2O$ 3c-DESs

ChCl, glycerol, and $AlCl_3{\cdot}6H_2O$ were mixed at specific molar ratios of 1 : 2 : (0.1, 0.13, 0.2, 0.28, 0.33), respectively, and



Scheme 1 Illustration of the design of the ChCl/glycerol/$AlCl_3{\cdot}6H_2O$ 3c-DES.

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stirred at 100 °C. The resulting transparent solvents were labeled as 3c-DES-1, 3c-DES-2, 3c-DES-3, 3c-DES-4, and 3c-DES-5.

Lignin fractionation

Three identical samples were mixed containing 1 g of poplar wood powder and 20 g DES, and the mixtures were heated in a 50 mL glass bottle at 110, 120, and 130 °C for 4 h each. The resultant dark brown liquid was poured in 200 mL acetone/water (70 : 30 v/v) solution and stirred for 2 h. The mixture was vacuum-filtered and washed with the same solution three times. After filtration, the mixture was then heated in a rotary evaporator at 60 °C to remove the acetone. Deionized water was added to the solution to precipitate the dissolved lignin.

DFT calculations

The relevant calculations were carried out with the Gaussian 09 program package. The initial geometry optimizations were obtained using the M06-2X exchange–correlation function combined with the 6-31G* basis set. All samples were subjected to re-optimization at the M06-2X/6-31+G** level. To eliminate the dispersion effect, we added the keyword of empirical dispersion = *gd*3 in the route section, and all calculations were optimized under no symmetry restrictions. Frequency analysis at the M06-2X/6-31+G** level was performed for the optimized structures to verify energy minima and provide zero-point energy corrections. The thermochemical characteristics were obtained for single-point energy calculations at the M06-2X/6-311++G** level to test their consistency and reproducibility.

The atoms in molecules (AIM) theory was employed to quantify the H-bonding interactions. Based on the AIM theory, the prominent criteria for the assessment of the existence of hydrogen bonds were proposed as follows: the presence of a bond path and an interatomic surface, the existence of the bond critical point (BCP), the range of ρ from 0.002 to 0.035 au, and the scope of $^{\Delta 2}\rho$ from 0.024 to 0.139 au.

The bond-acceptor–donor interactions were analyzed through 2D (the reduced density gradient *S*, *versus* the product of the sign of the second eigenvalue of the Hessian matrix and the electron density, signal $(\lambda_2)\rho$) and 3D-NCI (non-covalent interactions) plots. The 2D-NCI plot provides the information on the weak interactions and compares them with strong interactions. The 3D-NCI plot shows a three-color code to differentiate between attractive interactions (blue), very weak non-covalent interactions (green), and repulsive interactions (red). The darkest shade of blue corresponds to the strongest hydrogen bond.

Measurement of the K–T solvatochromic parameters

Nile red was used as a solvatochromic indicator to determine α, and 4-nitroaniline (NH_2) was used to determine β. *N*,*N*-Diethyl-4-nitroaniline (NEt_2) was selected to determine π^*. See Note S1 in the ESI† for details.

Fourier transform infrared (FTIR) spectroscopy

FTIR spectra were recorded using a Nicolet Magna 560 instrument (Thermo Fisher Corp., USA). The spectra were recorded in attenuated total reflectance mode, and the data were recorded from 650 to 4000 cm^{-1}, with 32 scans per spectrum and a resolution of 4 cm^{-1}.

Raman spectroscopy

Raman spectroscopy was performed with a Jobin Yvon HR800 micro-Raman spectrometer using an Ar^+ laser at 457.9 nm with 2 cm^{-1}.

^{27}Al nuclear magnetic resonance (NMR) spectroscopy

The samples were dissolved in 0.5 mL deuteroxide. ^{27}Al NMR spectra were obtained at 600 MHz (pulse duration, 11.59 μs; acquisition time, 0.26 s; 131 scans).

^{31}P nuclear magnetic resonance (NMR) spectroscopy

The samples were prepared by dissolving 40 mg lignin fractions in a 0.8 mL solvent mixture of pyridine/$CDCl_3$ (1.6 : 1). Then, 100 μL internal standard solution (400 mg cholesterol and 40 mg pyridine/$CDCl_3$) and 100 μL chromium acetylacetonate were added, followed by 130 μL 2-chloro-4,4,5,5-tetramethyl-1,3,2-dioxaphospholane reagent. ^{31}P NMR spectra were recorded with an Agilent Pro Pulse 500 MHz spectrometer operating at 202.3 MHz and using an inverse gated decoupling sequence and a relaxation delay of 5s. Approximately 1000 transients were acquired to ensure a high signal/noise ratio.

^{1}H–^{13}C 2-dimensional-heteronuclear single quantum correlation (2D-HSQC) NMR spectroscopy

A 90 mg sample was dissolved in 0.5 mL deuterated dimethyl sulfoxide (DMSO-d6). 2D-HSQC NMR spectra were obtained using an Agilent Pro Pulse 500 MHz spectrometer and recorded in the gradient HSQC adiabatic version. The spectral widths were 20 000 and 5000 Hz for the ^{13}C and ^{1}H dimensions, respectively. The number of collected complex points was 1024 for the ^{1}H dimension with a recycle delay of 1.5 s, the number of transients was 128, and 256 time increments were recorded in the ^{13}C dimension. Prior to Fourier transform analysis, the data matrices were zero filled to 1024 points in the ^{13}C dimension.

Gel permeation chromatography (GPC)

The molecular weights of the lignin samples were measured using a Waters 2695 GPC with a differential refractive index signal detector (Waters 2414). The column was operated at ambient temperature and eluted using tetrahydrofuran as the mobile phase at a flow rate of 1 mL min^{-1}.

High-performance liquid chromatography (HPLC)

The carbohydrates that dissolved in the 3c-DES were analyzed by HPLC after the removal of undissolved cellulose solids and dissolved lignin fractions. The glucose, xylose, and fructose contents were analyzed by HPLC (LC-1260, refractive index detector RID-G1362A, Agilent, USA). A Bio-Rad Aminex

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HPX-87P column (300 mm × 7.8 mm) was used with the following parameters: mobile phase: ultrapure water; flow rate: 0.6 mL min^{-1}; injection volume: 20 μL; column temperature: 80 °C; refractive index detector, retention time: fructose (10.8 min), glucose (11.4 min), and xylose (12.8 min). Furfural and 5-HMF were analyzed by HPLC (Waters 600E, UV detector VMD-2996, USA) using an Acquity BEH C18 column (1.7 μm, 2.1 × 50 mm), with the following parameters: mobile phase: acetonitrile : water (10 : 90, v/v); flow rate: 0.8 ml min^{-1}; injection volume: 20 μL; column temperature: 30 °C; UV detector: detection wavelength of 280 nm; and retention time: 5-HMF (8.4 min), furfural (13.9 min).

Results and discussion

Investigation of H-bonding interactions in ChCl/glycerol DES by DFT calculation

DESs are formed from mixtures containing several anionic, cationic, or nonionic species.[20] The properties of DESs vary with different HBA and HBD combinations. The different qualities of H-bonds lead to the difficulties associated with the quantification of H-bonding interactions.[21] Here, DFT calculation was employed to gain an insight into the H-bonding interactions of ChCl–glycerol at the atomic level. The molecular structures and optimized conformations of ChCl, glycerol, and the prepared ChCl/glycerol (1 : 2) DES are shown in Fig. 1. The choline cation (Ch^+) may be considered as a tetrahedron formed by the methyl/methylene groups of a quaternary ammonium head and a CH_2OH chain extending from one of the vertices and partially blocking the corresponding face (Fig. 1a). Choline can adopt either a *trans* or *gauche* conformation with respect to the N–C–C–O torsion angle. Additionally, three hydroxyl groups in the glycerol endow the ChCl–glycerol with numerous H-bond interacting capabilities (Fig. 1b). Thus, the H-bonding interactions in ChCl–glycerol (Fig. 1c) can be identified as: Ch^+ and glycerol (cationic H-bond), chloride ion (Cl^-) and glycerol (anionic H-bond), ChCl ion pairs (doubly ionic H-bond), and glycerol–glycerol (neutral H-bond). These interatomic distances are all within the accepted criteria of H-bond coordination (<3.5 Å).[22] The distance of Ch^+ and Cl^- delocalization is 0.397 Å, which is plausible to increase the degree of freedom of Cl^-.

The AIM theory was used to study the critical points between the HBD and HBA nuclei and quantify the H-bond interactions in terms of the electron density (ρ) and the Laplacian of the electron density ($^{\nabla 2}\rho$) at the bond critical point (BCP).[23] The BCP between two atoms is indicative of a bonding interaction, the ρ value describes the strength of a bond, and the $^{\nabla 2}\rho$ value indicates the characteristic of a bond (ionic bond, H-bond, or van der Waals bond). The criterion for the existence of an H-bond is a range of 0.002–0.035 au for ρ and 0.024–0.139 au for $^{\nabla 2}\rho$.[24] Fig. S1 in the ESI† shows a simulated topological graph of ChCl–glycerol, and Table S1† summarizes the values of ρ and $^{\nabla 2}\rho$ for all BCPs between the HBD and HBA. The positive values of $^{\nabla 2}\rho$ are all within the H-bond and van der Waals force range. The AIM molecular graphs in Fig. 2a show the strength of the anionic ($[Cl(glycerol)]^-$), cationic ($[Ch(glycerol)]^+$), and neutral (glycerol–glycerol) H-bonds in ChCl–glycerol. The O_1–H_7···Cl_{44} H-bond interaction with Cl^- is one of the strongest H-bonds, and ρ_{BCP} = 0.021 au, which is larger than that of an H-bond interacting mainly with Ch^+ (ρ_{BCP} = 0.014 au). Cl^- not only forms stronger H-bonds in Cl–glycerol, it is also circulated by van der Waals, C_{18}–H_{25}···Cl_{44} ρ_{BCP} = 0.005 au, and $^{\nabla 2}\rho$ = 0.014 au (Table S1 in the ESI†). Hence, the number and strength of anionic H-bonds are greater than those of cationic H-bonds, suggesting that the anionic H-bonds play dominant roles in ChCl–glycerol and manifold anions have actually been occupied by the H-bonds.

AIM was then visualized by employing the reduced density gradient analysis of the charges from electrostatic potentials to study non-covalent interactions. The scatter graph from the reduced density gradient analysis shows the corresponding H-bonding interactions, van der Waals bonding, and steric effects. The 2D and 3D non-covalent interaction plots in Fig. 2b reveal that the free sites of Cl^- are occupied due to the

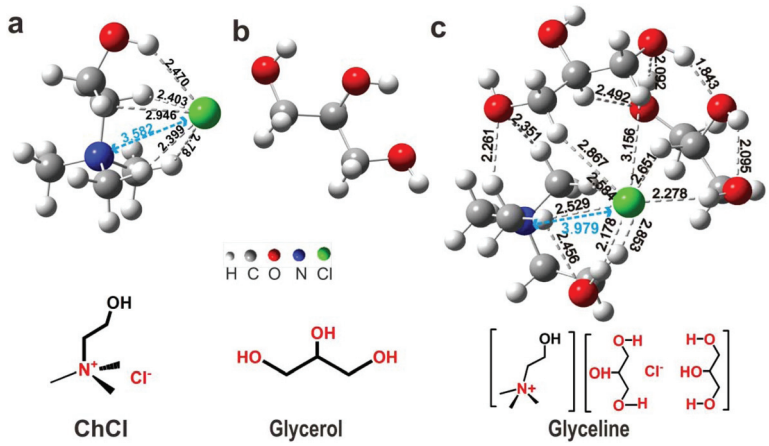


Fig. 1 Theoretical depiction of the interatomic distances of (a) ChCl, (b) glycerol, and (c) ChCl–glycerol (1 : 2) at the M06-2X/6-31+G** level of theory.

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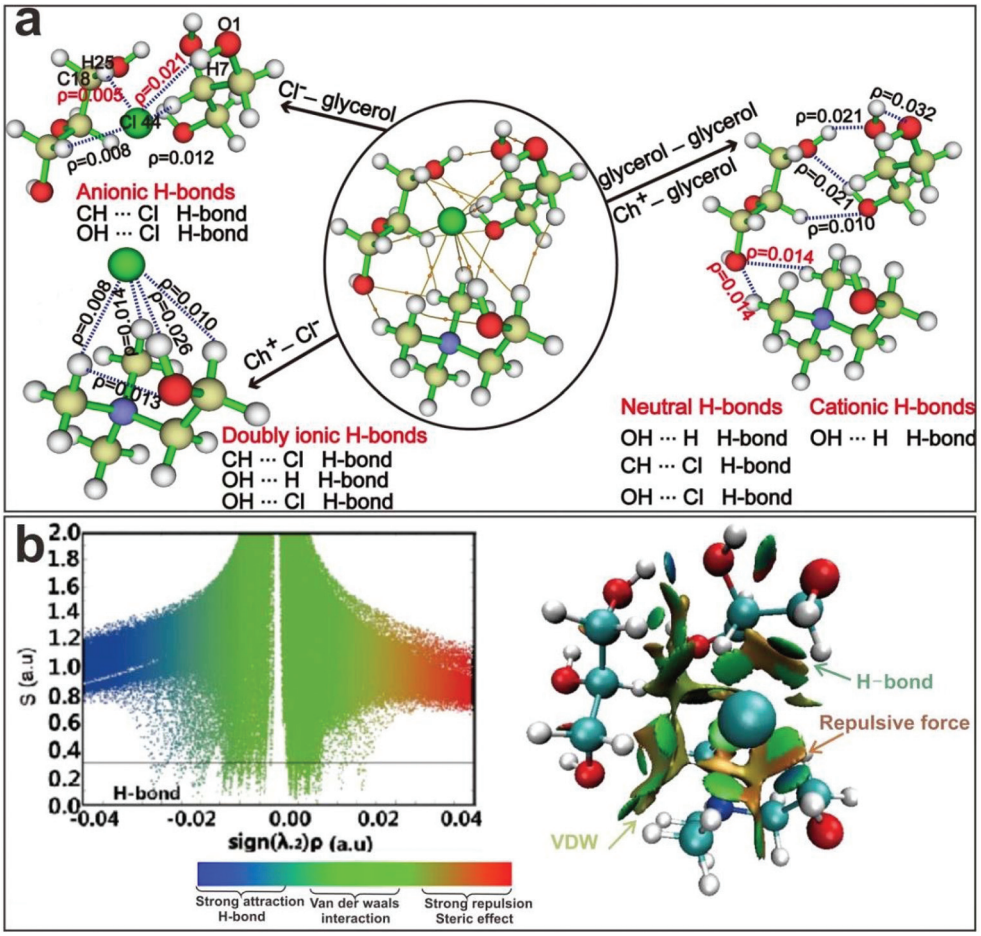


Fig. 2 (a) AIM molecular graphs of doubly ionic, anionic, cationic, and neutral H-bonds in ChCl–glycerol. (b) 2D and 3D non-covalent interaction plots of ChCl–glycerol.

coordination with glycerol to form different strength H-bonds and van der Waals bonds, which interfere with the ability of the initial constituents to crystallize and facilitate the formation of a liquid DES. Based on the above theoretical calculations, Cl^- was found to be surrounded by glycerol. The inefficiency of ChCl/glycerol DES for biomass fractionation is due to the insufficient HBA ability of ChCl–glycerol to compete with the inner H-bonds in LCCs.

We then calculated the H-bonding strengths in lignin using 1-(4-methoxyphenyl)-2-methoxyethanol as the model compound. Lignin has a variety of electron-rich phenylpropanols that are irregularly joined together through C–O and C–C linkages to create a complex structural network (Fig. S2, ESI†). The intramolecular H-bonds are mainly OH···O bonds (Fig. 3a). Most of the ρ values are close to 0.022 au (Table S1†). Thus, the H-bonding strengths in ChCl–glycerol are weaker compared to those in lignin (lignin–lignin > Cl^-–glycerol > Ch^+–glycerol), suggesting that ChCl/glycerol DES cannot break the H-bonds in lignin. Furthermore, the β–O–4 ether linkage is another common linkage in lignin, which accounts for approximately 50% of all linkages.[25] The bond dissociation energies (BDE) of phenethyl phenyl ether (a model of the β–O–4 ether linkage) for C–O and C–C in Fig. 3b were calculated as 62.25 kcal mol^{-1} and 76.22 kcal mol^{-1}, respectively, which are both larger than the H-bonding energy of 4.3–40 kcal mol^{-1}. This indicates that the H-bonding energy of the ChCl/glycerol DES was unable to cleave the ether bonds in lignin.

Design of 3c-DES using a multisite coordination strategy

Based on the above theoretical calculations, the inefficiency of ChCl/glycerol DES for biomass fractionation was ascribed to the weak H-bond-accepting ability of occupied-site anions and the inadequacy of protons/active acidic sites to cleave the H-bonds and ether bonds in LCCs. Therefore, inspired by the development of a Type IV DES (metal chloride hydrate with HBD),[26] seven types of metal chlorides/metal chloride hydrates were tested to mix with ChCl and glycerol, and a series of ChCl/glycerol/metal chloride solvents were formed (Fig. S3, ESI†). The DESs with hypervalent $AlCl_3{\cdot}6H_2O$ exhibited the maximal Lewis acid strength, the most prominent K–T parameters, and the strongest lignin fractionation ability (Table S2, ESI†). Hence, by increasing the amount of $AlCl_3{\cdot}6H_2O$ in ChCl/glycerol (1 : 2), five proportional ChCl/glycerol/$AlCl_3{\cdot}6H_2O$ 3c-DESs were formed. The resulting 3c-DESs all showed excellent

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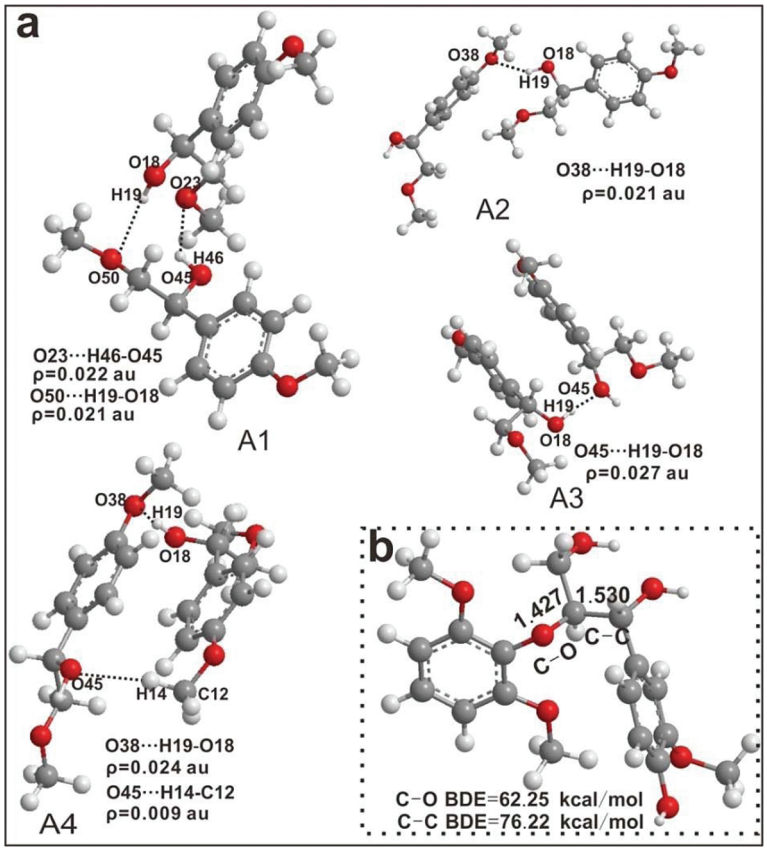


Fig. 3 (a) Optimized configurations for lignin–lignin calculated at the M06-2X/6-31+G** level. The corresponding atom numbering and H-bonds are marked. (b) The optimized structure of the dimeric lignin model. Bond dissociation enthalpies (C–O and C–C) were calculated at the M06-2X/6-311++G** level.

"deep" and "eutectic" characters at room temperature, and even at a low temperature of −20 °C (Fig. S4 in ESI†). Thus, this type of 3c-DES can be considered as a ternary DES.

The 3c-DESs were analyzed using FTIR and Raman spectroscopy. The force attenuation of intermolecular and intramolecular H-bonds in the ChCl/glycerol DES is illustrated by the decrease in the hydroxyl stretching frequency in the region of 3200–3300 cm^{-1} (Fig. 4a).[27] The emergence of a peak at approximately 1700 cm^{-1} was assigned to the stretching vibration of Al–H, and the signal intensity gradually increased with increasing $AlCl_3{\cdot}6H_2O$ content. The Raman spectra further reveal the characteristics of the 3c-DESs (Fig. 4b). The peak at 2903 cm^{-1} was assigned to $\nu_s(CH_2)$ and that at 2953 cm^{-1} was assigned to $\nu_{as}(CH_2)$,[28] which display enhanced asymmetric stretching vibrations once the 3c-DES is formed. This implies that the primary hydroxyls of glycerol are likely to coordinate with $AlCl_3{\cdot}6H_2O$ because aluminum's three-electron redox properties attract electron-rich oxygen.[29] From the ^{27}Al NMR spectra in Fig. 4c, the hydrated metallic salts as ligands can partly control the coordination chemistry around the aluminum center.[30] The resonance peaks in the 0–2 ppm range of the 3c-DESs originate from the octahedral hexa-coordinated aluminum species.[31] $[Al(OH)_2(H_2O)_4]^+$, which is considered the dominant species of $AlCl_3{\cdot}6H_2O$, can coordinate with glycerol as an unidentate ligand, and this correlation is ascribed to the HO···Al^{VI} coordination.[32] Al^{VI} is bound to glycerol by the oxygen atom in the form of $[Al(OH)_2(H_2O)_4]^+$ to form $[Al(OH){\cdot}(H_2O)_4{\cdot}(ligand)_n]^+$, leading to an upfield shift in the resonance (0–0.7 ppm).

The K–T solvatochromic method served as a quantitative tool to estimate the chemical properties of the 3c-DESs (Fig. S5, ESI†). K–T parameters α, β, and π^* represent the H-bond acidity, basicity, and dipolarity/polarizability of a solvent, respectively.[6,33,34] Based on the K–T parameters, the conductor-like screening model for real solvent prediction suggests that the high α and β values of DESs are crucial factors that correlate with the lignin extraction efficiency during biomass pretreatment. DESs with a high α value can easily donate H-bonds to liberate H^+, whereas that with a high β value has a high H-bond-accepting capacity. The α and β values of ChCl/glycerol DES were only 0.77 and 0.48, respectively (Table 1). In contrast, the α values of the 3c-DESs were improved to 1.9–2.0 (Fig. 4d); such high values indicate that it is very easy for them to release H^+. The β values also increased with the increasing $AlCl_3{\cdot}6H_2O$ addition and reached up to 0.68. The π^* values exhibited a slight decrease. The results fully prove that the acidic strength and H-bond accepting ability of the solvent increased after the incorporation of

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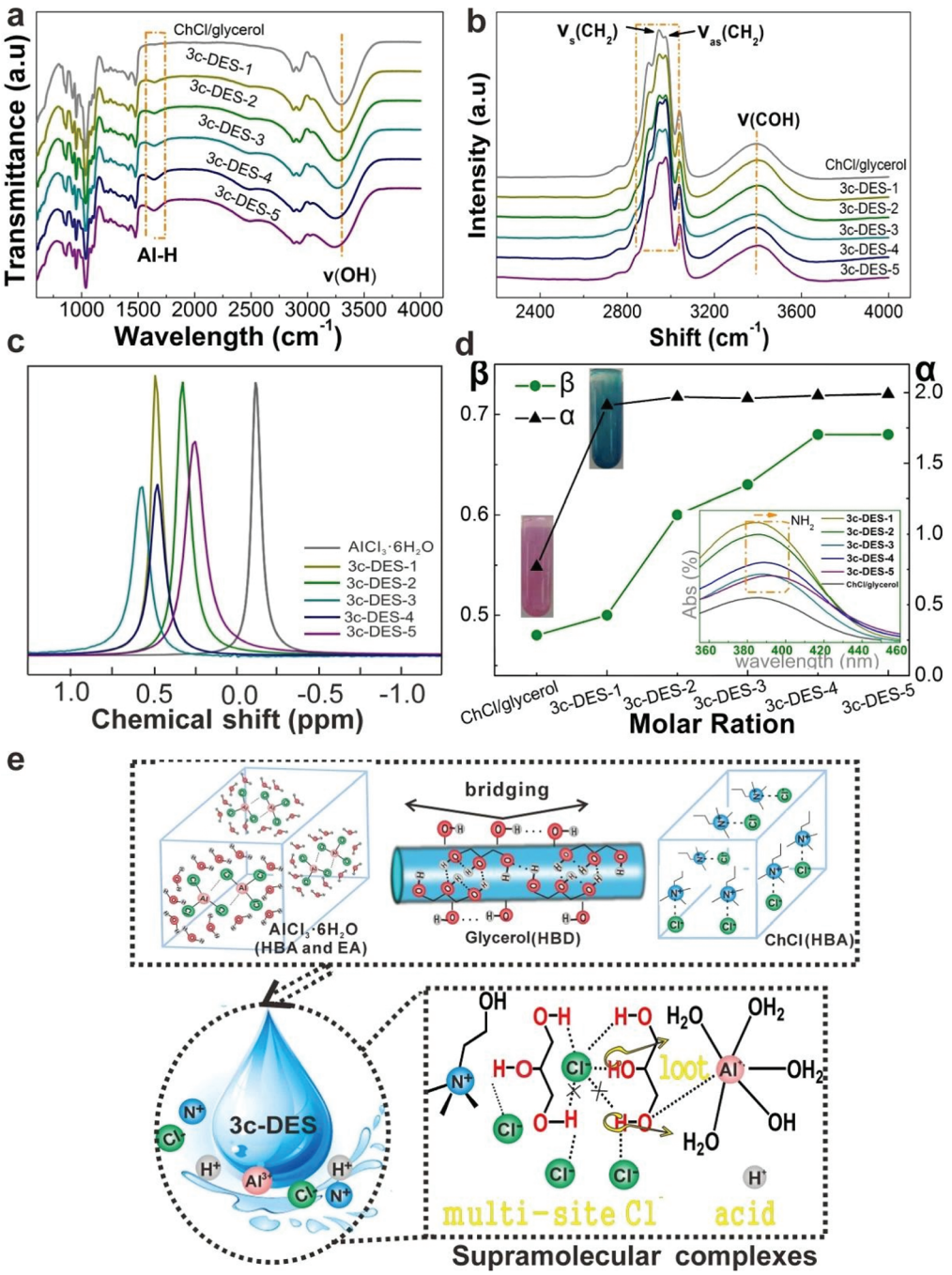


Fig. 4 (a) FTIR, (b) Raman, and (c) ^{27}Al NMR spectra of $AlCl_3{\cdot}6H_2O$ and the 3c-DESs with different $AlCl_3{\cdot}6H_2O$ ratios. (d) Changes of the α and β values in different 3c-DESs. (e) Schematic diagram elucidating the constitution principle of the 3c-DES system.

$AlCl_3{\cdot}6H_2O$. Therefore, the use of $AlCl_3{\cdot}6H_2O$ as an inexpensive additive agent in ChCl/glycerol can incorporate with the designable diversity of the DES and improve the latter's ability.

Fig. 4e depicts a schematic diagram of the construction of 3c-DES. First, apart from the anionic and cationic H-bonds in ChCl/glycerol DES, $AlCl_3{\cdot}6H_2O$ may attract the OH^- of glycerol to compete with ChCl, forming a larger number of stronger anionic ($OH\cdots Cl^-$) or neutral H-bonds ($OH\cdots O$), and simultaneously supplying multisite Cl^-. Second, the hexa-coordinated $AlCl_3{\cdot}6H_2O$ with electron-withdrawing substituents interacts with glycerol and forms unidentate aluminum ligands. Last, glycerol plays a "bridging role" by contributing OH^- groups to connect with ChCl and $AlCl_3{\cdot}6H_2O$ to form chlorine ion–metal cation–glycerol supramolecular complexes. The complex interactions significantly affect the entropy of the constituent materials, leading to a liquid ternary DES. The role of $AlCl_3{\cdot}6H_2O$ is deemed to cause substantial flexible interactions to form a complex coordinative system.

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Table 1 K–T parameter values of the ChCl/glycerol DES and the 3c-DESs

Constituents	Molar ratio	DES name	K–T parameters		
			π^*	α	β
ChCl : glycerol	1 : 2	ChCl/glycerol	1.18	0.77	0.48
ChCl : glycerol : $AlCl_3{\cdot}6H_2O$	1 : 2 : 0.10	3c-DES-1	1.19	1.91	0.50
	1 : 2 : 0.13	3c-DES-2	1.14	1.97	0.60
	1 : 2 : 0.20	3c-DES-3	1.13	1.96	0.66
	1 : 2 : 0.28	3c-DES-4	1.11	1.98	0.68
	1 : 2 : 0.33	3c-DES-5	1.11	1.99	0.68

Lignin fractionation using the 3c-DESs

Poplar wood powder was treated with the 3c-DESs by heating in an oil bath at 110, 120, and 130 °C for 4 h to verify their lignin fractionation efficiency (Fig. S6, ESI†). The mass percentages of the lignin fractions (LF) and cellulose residues (CR) were calculated (for the calculation methods, see Note S2 in the ESI†) and are listed in Table 2. The results show that the lignin extraction rate (LER) changes with the increasing amount of $AlCl_3{\cdot}6H_2O$ and increasing temperatures (Fig. 5a). The yield of LF increases from 0.69 ± 0.21% (using ChCl/glycerol DES treated at 120 °C for 4 h) to 18.2 ± 0.72% using 3c-DES-4 at 120 °C for 4 h, which accounts for 95.46 ± 0.82% of the lignin content in raw wood feedstock (Fig. S7, ESI†). The purity of the 3c-DES-4-extracted lignin (3c-DESL) reached 94 ± 0.45% (for the calculation method, see Note S3 in the ESI†). The 3c-DESL yield at 130 °C exceeds 19.12% (lignin content in raw wood feedstock); such a high value was possibly due to the generation of pseudolignin from carbohydrates (SEM image in Fig. S8 in the ESI†).[35] It is also worth noting that when $AlCl_3{\cdot}6H_2O$ and glycerol were used to form a Type IV DES (metal chloride hydrate and HBD), the LER could only reach 64.43% (Fig. S9 in ESI†); this result reveals that the H-bond interactions within the ternary DES are of great importance. Fig. 5b shows that the 3c-DESL yield also has a close correlation with the increasing β values, which confirms that high H-bond basicity can increase the capability of the solvent to disrupt the H-bond network.[36]

To explore the lignin fractionation mechanism of the 3c-DES, the interactions between Cl^- and the models of monomeric lignin were analyzed by DFT calculations (Fig. 5c). The results indicate that Cl^- is likely to form weak H-bonds with CH (CH···Cl, ρ < 0.015 au) and strong H-bonds with OH (OH···Cl, 0.028 < ρ < 0.036 au). The H-bonding strength of OH···Cl is higher than that of the OH···O bond in lignin, although O is typically considered as a good HBA. This proves that multisite Cl^- is required to compete with lignin to break the H-bond network in lignocellulose. From the previous BDE calculation in Fig. 3b, it is obvious that the weak ether bond (C–O) in lignin is most likely to be cleaved instead of the stronger C–C bond. The possible reaction pathway is shown in Fig. 5d. Firstly, acidolysis occurs in the 3c-DES to form a positive proton at C_α. After the deprotonation and protonation process, a resonance-stabilized positive charge is formed as shown in Fig. 5d(2). Finally, the ether bond is cleaved after the C_β is hydrated and deprotonated to form ketone and phenol groups.

2D-HSQC NMR spectroscopy was used to characterize the cleavage of the chemical bonds of 3c-DESL, and the milled wood lignin (MWL) was also characterized as a control (Fig. S10, ESI†). Ether units (A) and methoxyl groups (OMe) are the main linkages in lignin, and their specific signals are listed in Table S3 in the ESI.†[37] The typical syringyl-type ($A_{\beta\text{-G}}$) and guaiacyl-type ($A_{\beta\text{-S}}$) β–O–4 bonds in MWL lignin are illustrated in the figure. After the treatment, the signals correlating with A_α, $A_{\beta\text{-G}}$, and $A_{\beta\text{-S}}$ disappear from the 3c-DESL (Fig. S10b, ESI†), which proves that the 3c-DES successfully cleaved the β–O–4 compounds.[38] However, the typical C–C signals (β–β and β–5) still exist, which reveals that the 3c-DES did not cleave the C–C bonds in lignin. ^{31}P NMR spectroscopy analysis (Fig. S11, ESI†) indicates a significant increase of syringyl-type OH in 3c-DESL compared to that in MWL (Table 3).[39] This is due to the cleavage of lignin ether bonds. In addition, a relative decrease in the aliphatic OH of 3c-DESL occurs compared with that of MWL.

The weight-average molecular weight (M_w) and number-average molecular weight (M_n) values of MWL and 3c-DESL

Table 2 Mass percentage of cellulose residues (CR) and lignin fractions (LF) from different DESs, and the corresponding lignin extraction rate (LER, %)

Temperature	110 °C			120 °C			130 °C		
DES	CR	LF	LER	CR	LF	LER	CR	LF	LER
ChCl/glycerol	>90	≈0	≈0	>90	0.69	0.04	>90	0.71	≈0.04
3c-DES-1	61.34	11.89	61.29	61.07	14.58	75.15	64.46	17.31	89.22
3c-DES-2	52.90	12.89	66.44	51.87	17.04	87.83	62.86	19.10	98.45
3c-DES-3	46.69	15.34	79.07	46.11	18.12	93.40	52.05	20.37	105.00
3c-DES-4	46.50	16.23	83.66	42.01	18.52	95.46	42.68	20.41	105.21
3c-DES-5	45.65	16.21	83.57	42.43	18.45	95.11	38.44	20.42	105.26

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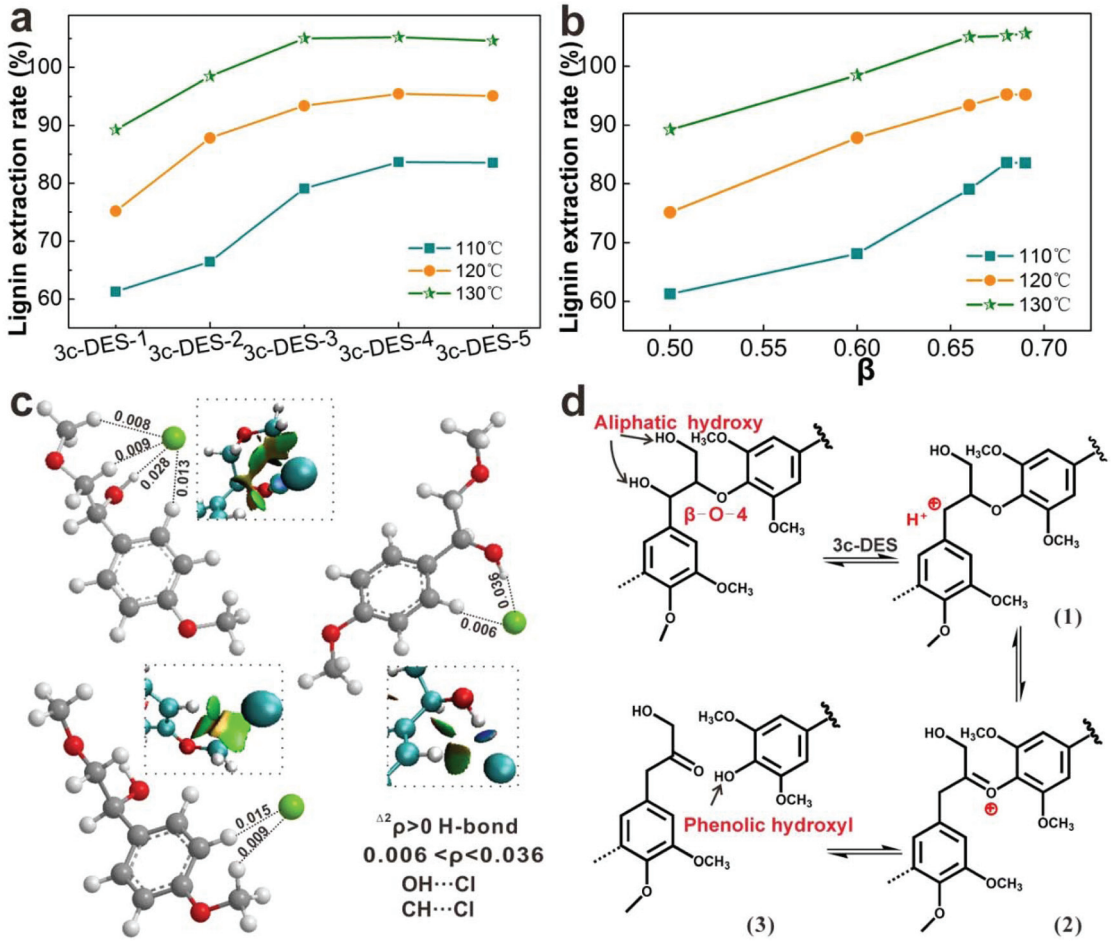


Fig. 5 Graphs showing (a) the changes of LER of the 3c-DESs at 110 °C, 120 °C, and 130 °C for 4 h, respectively; and (b) the correlations between the β values of 3c-DES-4 and the LER values. (c) The calculated geometries of lignin model-Cl^- complexes at the M06-2X/6-31+G** level. (d) Schematic diagram showing the possible reaction pathway of the cleavage of the β–O–4 compound.

Table 3 The hydroxyl group content of MWL and 3c-DESL

	δ (ppm)	MWL (mmol g^{-1})	3c-DESL (mmol g^{-1})
Aliphatic OH	145.4–150.0	3.913	2.125
All phenolic OH	137.6–144.0	1.113	1.583
Syringyl–OH	142.8–143.1	0.246	1.147
Guaiacyl–OH	140.0–138.8	0.440	0.268
Hydroxyphenyl–OH	137.0–138.2	0.427	0.168
COOH	134.5–135.5	0.168	0.184

were measured by GPC. As listed in Table S4 in the ESI,† the M_w and M_n of 3c-DESL are 5466 and 3270, respectively, and both are smaller than those of MWL. The polydispersity (M_w/M_n) is 1.67, which implies a homogeneous molecular structure and a narrow weight distribution of the 3c-DESL.

After removal of non-dissolved cellulose residues and the dissolved 3c-DESL from the solution, 30–40 wt% wood components still remained in the solution (Table 2). A carbohydrate analysis was then performed on the residual DES solution using HPLC. The results showed the presence of abundant saccharides including glucose, fructose, xylose, 5-HMF, and furfural in the solution (Table S5, ESI†). The amount of glucose only accounted for 8.6% of the original amount of glucose in the raw wood; 1.86 wt% fructose and 8.31 wt% 5-HMF were also detected. These amounts were probably detected because in the 3c-DES, some part of the amorphous cellulose hydrolyzed into glucose, and glucose was then isomerized to fructose and dehydrated to 5-HMF. 5.12 wt% xylose was detected from the solution (accounting for 25.6% of the xylose in the raw material) and also 1.86 wt% fructose. These saccharides are deemed to have been converted from hemicellulose and amorphous cellulose through DES hydrolysis. These results hint that DES is also promising for carbohydrate valorization.

Conclusion

This work aimed to explore why the ChCl/glycerol DES does not display a powerful lignin fractionation efficiency during biomass pretreatment; quantum mechanics calculations, spec-

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troscopy characterization and K–T solvatochromic parameters were used. DFT calculation was reasonably used to understand the molecular interactions within the constituent mixtures of ChCl/glycerol DES. An inefficiency mechanism was found due to the weak competing interactions of ChCl–glycerol with the linkages in the LCCs. The Cl^- in ChCl–glycerol was surrounded by mutually anionic H-bonds ($[Cl(glycerol)]^-$) and cationic H-bonds ($[Ch(glycerol)]^+$), which resulted in the weak H-bond-accepting ability of occupied-site anions and inadequate protons/inactive acidic sites. For this reason, the K–T parameters of ChCl/glycerol DES, especially α and β, were significantly smaller than those of powerful DESs. In this regard, a molecular design of a ternary DES was proposed based on an acidic multisite coordination strategy. $AlCl_3{\cdot}6H_2O$ as an anion donor and active acidic site holder was introduced in ChCl/glycerol. The competition of anionic H-bonds (CH···Cl/OH···Cl) and unidentate aluminum ligands ($[Al(OH){\cdot}(H_2O)_4{\cdot}(ligand)_n]^+$) was synchronized to form chlorine ion–metal cation–HBA supramolecular complexes. The resulting 3c-DES showed typical "deep" and "eutectic" characteristics. Consequently, the K–T parameters α and β of the 3c-DES significantly increased to 1.99 and 0.68, respectively. The lignin extraction rate using the 3c-DES was improved to 95.46% with a purity of 94%. The reduction of molecular weight and polydispersity ensured that the LF has good characteristics for further utilization. In conclusion, this work presents a contribution to the general areas of sustainability of biomass refinery with a specially designed 3c-DES. Such work may also inspire rising research enthusiasm in the exploration of DESs tailored for bio-refinery and bio-transformation.

Abbreviations

AIM	Atoms in molecules
BCP	Bond critical point
BDE	Bond dissociation energies
ChCl	Choline chloride
DES	Deep eutectic solvent
DFT	Density functional theory
H-bond	Hydrogen bond
HBA	Hydrogen bond acceptor
HBD	Hydrogen bond donor
K–T	Kamlet–Taft
LCC	Lignin–carbohydrate complex
MWL	Milled wood lignin
3c-DES	Three-constituent DES

Author contributions

H. Y. conceived the idea and supervised the project. H. Y. and Q. X. designed the research. Q. X. carried out most of the experiments and analyzed the data. Y. Z. L., J. M. and W. C. participated in the experiments. H. Y., Q. X. and Y. Z. L. discussed the results and prepared the manuscript. W. C., S. L., Y. X. L. and J. L. provided some useful suggestions.

Conflicts of interest

There are no conflicts to declare.

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